Fairness Has Memory

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Abstract

When a scarce, indivisible good is allocated by lottery among people with equal claims, the standard theory of fairness says the lottery is the fairest thing available. When the same good is allocated the same way year after year to largely the same people, the theory has nothing further to say: each lottery is fair, so the sequence is. This is a mistake. A sequence of individually fair lotteries can be an unfair allocation, and where claimants stay in the pool after being served, fairness requires that their chances in later rounds depend on the history of their unsatisfied claims. The first two claims use only premises the standard theory accepts: that a divisible good must be divided in proportion to claims, and that a lottery is a second-best surrogate for division. Over a sequence, the good becomes divisible. I then show what memory should track, units where later units are as good as earlier ones and waiting where they are not, and how heavily it may press: only through a rule whose transfers between classes of claimants are a by-product of a past winner's not being taxable, never its mechanism, and only up to the weight at which a claimant is excluded when she holds a whole unit of another's share. Where the good recurs, rules that forget and rules that remember lexically are unfair in opposite directions; where winners leave the pool and some will leave unserved, the lottery that forgets is what fairness requires.

1 Two lotteries

Every year Nevada and New Mexico have more people who want to hunt mule deer than tags to give, so each holds a lottery. Nevada remembers: an unsuccessful applicant earns a bonus point, her points are squared to fix how many random numbers are drawn for her the next year, and her lowest number is her entry, so ten years of losing gives her a hundred and one draws against a newcomer's one.[1] New Mexico forgets: "New Mexico does NOT grant preference to applicants who were unsuccessful in previous drawings."[2] A hunter passed over for ten years enters the eleventh draw as a first-time entrant.

Many institutions side with Nevada, though they disagree about how: Arizona adds one random number per unsuccessful year and reserves a fifth of its tags for the highest point-holders; Colorado gives most licences to whoever has the most preference points, chance only breaking ties. The Grand Canyon's river-permit lottery weights by years since an applicant's last trip, capped at five, and starts a newcomer at the cap.[3] Others forget, among them the lotteries that allocate some 85,000 H-1B visas and up to 55,000 Diversity Visas a year: a worker not selected in three successive years enters the fourth draw as a newcomer.

The theory of fair allocation has barely touched this disagreement, and where it does, sides with New Mexico. The best-developed account, John Broome's (1984a, 1991a, 1991b), holds that fairness requires claims to be satisfied in proportion to their strength, and that when the good is indivisible and claims are equal, an equal lottery is the fairest

[1] Nevada Department of Wildlife, "Bonus Point Program," https://www.ndow.org/blog/bonus-point-program/.

[2] New Mexico Department of Game and Fish, "How the New Mexico Draw Works," https://wildlife.dgf.nm.gov/hunting/applications-and-draw-information/how-new-mexico-draw-works/.

[3] National Park Service, "Weighted Lottery," https://grcariverpermits.nps.gov/faq.cfm.

available procedure. It evaluates a lottery on its own, and on its own each year's New Mexico draw is impeccable. Broome's critics, who locate the fairness of lotteries in what they express or exclude rather than in the chances they distribute (Wasserman 1996; Stone 2007, 2011; Spiekermann 2022), reach the same verdict: an impartial procedure is fair whatever last year's did. The wider literature takes the single lottery as its unit (Sher 1980; Kornhauser and Sager 1988; Elster 1988; Duxbury 1999; Saunders 2008; Stone 2008). The nearest to my thesis is a page of Hersch and Rowe's (2025, 3): "many goods can be divided temporally as well as spatially," so that an indivisible good can be divided "over time through a rotation," which they credit to Elster (1989), and for recurring chores "who completed the chore last time matters to who has a claim to avoid it this time." That is the divisibility argument in miniature. In computational social choice (Lackner 2020; Lackner and Maly 2023; Freeman et al. 2017; Benade et al. 2018) history-dependence is an axiom. What follows adds the derivation inside Broome's theory, against his own extinction thesis; the constraint on how hard memory may press; and the structure in which it is forbidden.

I will argue for three claims. *Non-separability*: the fairness of a sequence of allocations is not a function of its members' fairness as the standard theory assesses it. *Memory*: when units of a good are allocated repeatedly among claimants who remain in the pool after receiving one, fairness requires that each claimant's chances in later rounds depend on her history of unsatisfied claims. *Currency and weight*: what memory tracks is a claimant's shortfall from her fair share, in units of the good where a later unit is as good to her as an earlier one and in waiting where it is not; and it may press that shortfall only through a rule whose transfers between classes of claimants are a by-product, not its mechanism, and only up to the weight at which a claimant is excluded when she holds a whole unit of another's share.

2 Claims, chances, and division

A claim, in Broome's sense, is a duty owed to the candidate herself that she should have the good, as distinct from other reasons for giving it to her, such as usefulness (Broome 1991a, 91–93). Fairness requires that "equal claims require equal satisfaction," that "stronger claims require more satisfaction than weaker ones," and that "weaker claims must not simply be overridden by stronger ones" (Broome 1991a, 95).

Two features of the theory carry my argument. The first I will call the *Division Principle*: where a good is divisible, proportional satisfaction is achievable and required: if two people have equal claims to some good, it "should be divided," even when giving it all to one of them would do more good (Broome 1991a, 95). Broome states it most generally in a passage that already divides across time:

> The good, for instance, of avoiding military service could be divided by conscripting everyone for a short time rather than a few people for a long time. If people's claims to a good are equal, then fairness requires that the good should be divided between them if it can be. That is the only way to treat them with perfect equality. (Broome 1984a, 47)

The second is the account of lotteries as second best. When the good is indivisible and scarce, a lottery gives "a sort of partial equality in satisfaction": "Each person can be given a sort of surrogate satisfaction. By holding a lottery, each can be given an equal chance of getting the good. This is not a perfect fairness, but it meets the requirement of fairness to some extent" (Broome 1991a, 97–98). Broome grants this only if a chance counts as surrogate satisfaction, which he finds plausible (98). Two things follow. Where division is available, it is required. Where it is not, the surrogate is owed: a claimant who cannot be given her portion must be given her chance.

What duty does Nevada owe anyone to let her shoot a deer? Broome declines to say which reasons are claims,

beyond granting that some are (Broome 1991a, 93). The claims in my cases arise from the allocator's undertaking: a body that announces it will allocate a public resource fairly among eligible applicants thereby owes each of them fair treatment, and what that requires is not the agency's to stipulate. New Mexico's announcement that it grants no preference to past losers describes its procedure, not its fairness.

3 The divisibility argument

Consider a good that comes in indivisible units, one per period, allocated by one agency among equal claims, where receiving a unit does not remove a claimant from the pool: a tag a year, a night off the rota a week. Suppose for now that the same N claimants are present for T periods. Within any one period the standard verdict is correct: one unit, N equal claims, no division, so an equal lottery. But the units are indivisible and the sequence is not. Over T periods it is possible to give each claimant T/N units, or as near as the integers allow. Call the procedure that does so *rotation*: draw a random order once and cycle through it. Rotation is, in the sequence, what dividing the food is in the single case. The sequence can be divided, so fairness requires rotation, or whatever comes as close to it as circumstances allow.

Compare New Mexico's procedure, a fresh equal lottery every period: the *forgetting* rule. Over T periods it does not divide the sequence but raffles it. With ten claimants and ten periods it leaves, on average, three and a half with nothing while others receive two or more.[4] Rotation leaves nobody with nothing. The forgetting rule does over the sequence what the theory forbids in the single case: it takes a divisible good and gives it out by lot.

As a reductio: suppose someone proposed to allocate a cake among ten people with equal claims by cutting it into ten slices and raffling each slice separately. Each raffle, taken by itself, is what the theory recommends for an indivisible good. Nobody thinks this a fair way to allocate a cake; the theory itself says the cake should be divided, which means giving everyone a slice, not a chance at each. A cake handed out over ten days is still a divisible cake.

The first consequence is non-separability. The standard theory assesses a lottery by the claims present when it is held, fixed by need, desert and the like, not by earlier lotteries. Call a rule *locally fair* at a period if its procedure then, so assessed, is fair. The forgetting rule is locally fair at every period; rotation at none after the first, since given the history one claimant is certain of the unit. Yet rotation is the fair allocation of the sequence and the forgetting rule is not. So fairness, so assessed, is not separable across occasions. Put another way, as I will from §4 on: if claims carry history, so that a claimant who has received less than her share has a stronger claim to the next unit, then rotation *is* locally fair at every period and fairness is separable after all. These are one thesis in two vocabularies; what is not available is history-blind claims *and* separable fairness.

The second consequence is memory. Any rule that divides the sequence must give units to those who have received fewer, which requires knowing who they are.

The argument assumes something an opponent will press. The Division Principle applies to a good that "could be divided," and the tags a hunter draws in different years are not portions of one tag. What licenses treating them so is *Fungibility*: units allocated in different periods are portions of one divisible good, for a claimant, to the extent that her claim is to a unit of that kind and a later unit would satisfy it as an earlier one would. The hunter's claim is to a tag, not to the 2027 tag. Fungibility is a matter of degree, since delay is nearly always some cost, and it fails outright when a later unit is much worse for the claimant: the visa applicant whose sponsor will not wait. Then the claim is largely to timeliness. But timeliness is divisible: delay can be shared out as units cannot, so the argument changes

[4] The expected number with nothing is $N(1-1/N)^T$. The simulation code behind every figure in this paper accompanies it.

currency. Fairness requires that the *waiting* be divided in proportion to claims, where the constraint of §5 allows it (§5.1).[5]

4 The extinction objection

The argument seems to run into a doctrine Broome states in the same year. Once a fair lottery has given Michael the dialysis machine, it would be absurd to hold another on the ground that lotteries are fair: "once Michael has won the first lottery he has a perfect claim to the treatment, and Maggie has none. Consequently it is not at all fair to hold a second lottery" (Broome 1984b, 628–29). Call this the *extinction thesis*: losing a fair lottery extinguishes the loser's claim. Vong (2015) keeps the thesis in one form: a *procedural* claim, once satisfied by a fair lottery, is satisfied, while a *benefit* claim "cannot be lost due to the results of a procedurally fair lottery" (pp. 479–480); so when new units arrive the losers are "a new case of scarce benefit distribution" and get a fresh fair lottery (p. 483). In Vong's cases, where recipients leave the pool, that is a fresh lottery among those who did not benefit, which is the forgetting rule; and where the literature does give a lottery memory (Feldblyum Le Blevennec 2023), it is the *winner's* standing that is protected when new claimants arrive.

The reply distinguishes three structures the objection runs together. A *re-run*: the same claimants, the same unit, a second lottery because the first went the wrong way for someone. Broome is right that this is absurd, because there is nothing left to divide. An *interrupted allocation*, Vong's case: one unit, and new claimants arriving before it is handed over. And *recurrence*: a new unit each period, among overlapping claimants. Here the extinction thesis is true and irrelevant. Maggie's claim to the 2026 tag was extinguished when she lost; it is someone else's. But there is a 2027 tag, and the Division Principle says the two together are a divisible good, so the 2027 tag goes, or is more likely to go, to someone who did not get the 2026 one. It is a claim to a unit that exists, with more force than a past winner's for the same reason that, when a cake is divided, someone who has had no slice has more of her claim outstanding than someone who has had one. Where the currency is waiting and everyone will be served, the same holds with waiting in place of slices.

To bite, Broome's objection needs a stronger thesis: that losing a fair lottery for one unit extinguishes any bearing the loss might have on later units. That is what the cake refutes: someone who raffled each slice could say of each raffle that the losers' claims to that slice were fairly extinguished, and the cake would still not have been divided. Vong's form fails for a related reason: a procedural claim is satisfied by a fresh equal lottery only if a fresh equal lottery is the fair procedure, and over recurring units the Division Principle says it is not. Vong rejects views on which losing weakens a claim and on which winning strengthens it (pp. 477–479); extended to recurrence, his view that a benefit claim survives a loss is an ally of mine, and the question his objections, about new claimants for a single unit, do not reach is whether winning weakens a claim to *later* units.

A residual worry makes the surrogate character of chances do work. If the 2026 lottery gave Maggie surrogate satisfaction, does extra weight in 2027 not satisfy her twice? A chance is not a portion of the good but what one gives when a portion cannot be given. If an equal chance of a slice were a kind of equal slice, the raffle of the cake would be a division of the cake, and the theory could not say that the cake should be divided instead. What memory counts is units received, not chances.

[5] With unequal claims the problem is apportionment (Wintein and Heilmann 2018), whose divisor methods have a sequential form (Balinski and Young 2001, 100, Prop. 3.3). "Repeated" fair division (Igarashi et al. 2024; cf. Aleksandrov and Walsh 2020) concerns envy, not claims.

5 What memory tracks

Sort the institutions by structure. In *recurrence* a claimant who receives a unit stays in the pool: tags, permits, chores. In a *bottleneck* she leaves: a visa, a flat, a kidney. (Hersch and Rowe use "bottleneck" more narrowly, for queues everyone eventually clears.) Structure is fixed by whether winners leave and currency by Fungibility, and in practice they run together, since whoever leaves on being served was waiting for one unit and cared when it came.

5.1 Shortfall in units

In recurrence the argument of §3 fixes what memory tracks: the difference between the share of units a claimant should have received by now and the number she has, her *shortfall*. With equal claims and one unit per period, a claimant present for r periods who has received w units has shortfall $r/N - w$. Shortfall is neither the number of losses nor time waited: points-for-losses systems reset on a win, treating a claimant who won once in twenty years as a newcomer, and count a loss among two hundred applicants as one among two; time waited tracks shortfall only if units and claimants are constant in number.

Consider the rules that press shortfall with a weight $\kappa \geq 0$: each claimant's chance is $1/N$ plus κ/N times the excess of her standing over the pool's mean, floored at zero, where standing is shortfall kept relative to the pool: a newcomer enters level with the pool's mean, and a claimant who leaves takes her standing with her (§6). At $\kappa = 0$ this is the forgetting rule. At $\kappa = 1$ a claimant's distance from the mean shrinks by about a factor of $1 - 1/N$ each period, so it is corrected over roughly the time the practice takes to hand out one round of shares. A past winner's chance falls below a newcomer's, and reaches zero once she is $1/\kappa$ units ahead of the pool; she is then excluded until the others catch up, but never taxed. As κ grows the rule approaches giving each unit to whoever is furthest behind, which is Aleksandrov et al.'s (2015) Balanced Like. In a fixed population everyone starts level, so each has the same chance of every unit whatever κ is, while dispersion falls: its variance is $\frac{N-1}{2\kappa N}$ absent clipping, and with ten claimants and ten periods the expected number left with nothing falls from 3.5 under the forgetting rule to 2.1 at $\kappa = 1$, 1.4 at $\kappa = 2$ and 0 at $\kappa = 10$. In a fixed population the Division Principle says: press as hard as you can, and rotate.

The rule has near relatives. Lackner's (2020) perpetual voting rules weight voters by the history of past decisions, since "only due to this dependency we can achieve fairness over time" (2104); his Perpetual Quota weights each by $\max(0,\ \text{quota} - \text{satisfaction})$ (2106), which in my setting is shortfall. But his rules are deterministic, so here Perpetual Quota *is* rotation in its tie-breaking order, and the memoryless lottery enters only as a benchmark (2109). The lottery with memory lies between, and the theory of claims requires it, since an arbitrary fixed order among equal claims is what it forbids. Rules weighting by losses alone fail even simple proportionality (Lackner 2020, Prop. 1; Lackner and Maly 2023, Thm. 2), a deterministic analogue of the failure of points-for-losses.

5.2 Newcomers and the surrogate constraint

Once claimants come and go, a real objection appears. A newcomer has as much of her claim outstanding as any past loser had on first application, and may be gone before the division reaches her; under any rule with memory she is given a smaller chance than that loser now has.

The answer also fixes how heavily memory may press. Division of the sequence is available only to a claimant who stays long enough for it to be completed; one who leaves after three periods cannot be given three-tenths of a tag; for her the sequence is as indivisible as the unit. If the allocator knew who would leave, it would owe them the surrogate outright: a chance, each period, proportional to their claim. Not knowing, it owes the surrogate to everyone, in

the only form available: that no claimant's expected share, over whatever tenure she turns out to have, fall short of proportionality. Call this the *surrogate constraint*. It is measured over the chances the rule assigns, exit taken as given; a claimant who leaves because she lost has a complaint against her luck, not the rule. The chance is what a claimant who will not stay for the division is *owed*, and a rule that takes it from her to smooth the shares of those who stay overrides her claim, whereas the dispersion it would smooth is the residue of a lottery, which the standard theory already *tolerates*. That ranking is mine, not Broome's, though it rests on his distinction between a claim overridden and a claim met in part; and since the surrogate is owed to each claimant who may leave, not to a class, their number does not bear on whether it is owed.

Three things hold of this constraint when exit is independent of history. First, it can only be met exactly: chances sum to one in every period, so under any rule the pool's expected share is proportional, and no tenure class can stand above proportionality unless another stands below it. Memory may redistribute realized shortfall across tenures, never expected share. Second, the shortfall rule meets it, for every κ, so long as no chance is clipped at zero. A newcomer enters level with the pool, so her chance in her first period is $1/N$ whatever the pool's state; thereafter her expected distance from the mean shrinks each period by the factor $1-\kappa/N$ and gains no constant term, so it stays zero, and every tenure class receives exactly its share. Third, what memory can do has a closed form. With exit rate q the variance of shortfall at exit is about $\frac{N-1}{N(Nq+2\kappa(1-q))}$, so the reduction relative to forgetting depends on κ and on the number of units a claimant can expect to receive during a stay, $1/Nq$. Where a claimant can expect one unit during her stay, as in Table 1, $\kappa = 1$ removes over a third of the dispersion; where a hundred claimants compete for each unit and stay ten years, under a tenth; as units per stay vanish, the verdict of this section shades into that of §5.1.

Clipping breaks the exactness, and no weight escapes it: at any $\kappa > 0$ a claimant far enough ahead of the pool wins with positive probability, her chance cannot fall below zero, and her unpaid share then falls on everyone, the newcomer included. So the constraint, read as a floor no class may ever fall below, admits no memory under churn, only forgetting. That reading mistakes what the floor protects. What the newcomer is owed is that her chance not be the *mechanism* by which the stayers' shares are smoothed. Under the shortfall rule it is not: the smoothing is done within each claimant's own stay, by moving her chance from periods when she is ahead to periods when she is behind, and the leak to other classes is the by-product of a boundary, her not being taxable; without the boundary it is zero at every κ, and as κ falls it shrinks faster than any power of κ while the smoothing shrinks only in proportion. Under points, queues and years-since-a-win the transfer from short tenures to long is the mechanism itself: it does not vanish as the weight falls, and there is no boundary to blame. That is the difference in kind the constraint marks, and it separates recurrence from the bottleneck of §5.1, where nobody present has received a unit, so that any smoothing of waiting is transfer between claimants and nothing else. It fixes the kind of rule; the unit fixes the weight. A claimant's chance reaches zero when she is $1/\kappa$ units ahead of the pool, and the Division Principle counts in units; so at $\kappa = 1$ exclusion begins exactly when she holds a whole unit that is someone else's share, a larger κ excludes for fractions of the indivisible unit, and a smaller one leaves a chance with someone who already holds another's unit. Table 1 shows the sizes.

Table 1: Ten claimants with equal claims, one unit per period, each leaving with probability 0.1 per period and replaced; 200 runs of 1,500 periods. A newcomer's chance in her first period and expected shares at longer tenures, as multiples of the proportional $1/N$ per period; standard errors at most .003 in the first column and .02 elsewhere. Dispersion is the standard deviation, in units, of a claimant's shortfall when she leaves.

Rule	Newcomer's chance	Expected share, tenure 10	Tenure 30	Dispersion of realized shortfall
Forgetting ($\kappa = 0$)	1.000	0.99	0.97	0.95
Shortfall, $\kappa = 1/2$	1.000	0.99	1.02	0.71
Shortfall, $\kappa = 1$	0.993	1.00	1.00	0.60
Shortfall, $\kappa = 2$	0.943	1.00	1.01	0.51
Shortfall, $\kappa = 10$	0.523	0.97	0.99	0.42
Points for losses, linear (Arizona, random pass)	0.263	0.96	1.15	0.69
Points for losses, squared (Nevada)	0.082	0.99	1.21	0.58
Queue / preference points (Colorado)	0.000	1.01	1.26	0.47
Years since last win, capped (Grand Canyon)	1.169	1.00	0.99	0.80

The leak costs the newcomer three hundredths of one per cent at $\kappa = 1/2$, under one per cent at $\kappa = 1$, six per cent at $\kappa = 2$ and nearly half her chance at $\kappa = 10$, against reductions in dispersion of a quarter, over a third, nearly a half and over a half.

The rules in use breach the constraint outright. Linear points give a newcomer a quarter of the chance her claim warrants and a claimant who stays thirty periods 15 per cent more than her share; squared points cut the newcomer to a twelfth; a queue gives her nothing, and whoever leaves early leaves with nothing. These rules reduce dispersion, some by more than the moderate shortfall rules, but transfer from short tenures to long is their mechanism, which denies the short-stayers the surrogate they are owed. The Grand Canyon's rule errs the other way: starting a newcomer at the cap gives her a sixth more than her share, which by conservation the others pay for.

So, in recurrence, the third claim is established. Memory is required, through a rule whose transfer between classes is the by-product of non-taxability rather than its mechanism; the rules in use, whose mechanism it is, fail, and to remember lexically lets an incumbent's accumulated claim override a newcomer's present one. Within the kind, the unit fixes the weight: at $\kappa = 1$ a claimant is excluded exactly when she holds a whole unit of another's share, every tenure class stays within one per cent of its own, and over a third of the dispersion goes. The residual lottery is the surrogate the short-stayers are owed. Only where the pool is fixed is it zero, and there the rule is rotation.

5.3 Bottlenecks and waiting

In a bottleneck nobody accumulates more than one unit; what varies is time waited. Fungibility fails, because each claim is to one unit and to having it soon; what is divisible is the waiting. Under the forgetting rule, waiting is

raffled: with a pool of ten kept full by arrivals and one unit per period, about one in seventy waits forty periods or more, though everyone's expected wait is nine. A queue serves everyone at nine. Time, unlike a flat, is divisible, so the Division Principle applies to the waiting and says: equalize it. This is why John and Millum (2020, 181, 205–6) are right that a queue can be as fair as a lottery, and Hersch and Rowe (2024) right to separate "bottleneck" cases, where queues belong, from "scarcity" cases, where lotteries do; between them lies recurrence, where a queue gives a short-stayer nothing.

The surrogate constraint carries over, and here it cuts the other way. Where claimants may leave unserved, by dying or losing eligibility, and the allocator cannot tell who, those to whom the division is not available are owed the surrogate: an equal chance, each period, at the unit. A pure queue denies it outright; so, to a degree, does any rule that weights by time waited. In recurrence the transfer between classes was the by-product of a boundary; absent clipping the shortfall rule gives every tenure class its share. In a bottleneck there is no boundary to blame: nobody present has received a unit, so the shortfall rule itself reduces to forgetting, and a rule that weights by waiting takes from the newcomer by its mechanism, whatever the weight. So the result is two-sided a second way. Where everyone will be served, the waiting is divisible and fairness requires a queue; where some will leave unserved and the allocator cannot tell who, fairness requires the lottery that forgets. Memory is required in one structure and forbidden in the other; the difference is whether a claimant's own stay has periods to move a chance between. Housing lists and transplant lists, where some always leave unserved, fall on the second side so far as exit is unforeseeable. Where it is foreseeable, or rises with waiting, the claim is urgent, time waited is evidence of urgency, and the rule must weight it, as transplant allocation does: a difference in claims, not a memory of history. (Bighorn ram tags, which New Mexico issues once in a lifetime,[6] are a bottleneck; there New Mexico's rule is right.)

The verdict for the visa lotteries follows, and it is not what the opening suggests. They are bottlenecks with attrition: a selected worker leaves the pool, and an unselected one may lose her sponsor or status unforeseeably. So the lottery that forgets is the fair rule, and Hersch and Rowe's (2024, 188) endorsement of the Diversity Visa lottery is right, though for a different reason: the surrogate owed to those who will leave unserved cannot be paid by anyone else. And their remark that a periodic lottery in a bottleneck is unfair because a person who "arrives into the lottery pool early and is continuously unlucky" may wait far longer than a lucky newcomer (199) holds only where everyone will be served.

6 How far does memory reach?

If Nevada must remember Maggie's 2026 loss in 2027, must it remember her loss in the housing lottery? In Arizona's draw? A view that cannot stop somewhere principled collapses into a demand that luck be equalized, a more contested thesis. Two bounds, besides the bounds of kind and weight in §5, follow from the argument that generates the claim.

Memory is bound to the practice. The corrective claim arises because a particular agency's allocation of a particular good left residual unfairness: a claim on *those* units, addressed to *that* agency. The good matters because the 2027 tag can divide the sequence only because it is a tag; a flat is no portion of what the hunter has a claim to. Goods are individuated by claims, not names: a Nevada tag and an Arizona tag are different goods, since a claim to hunt in one state is not satisfied by a tag for the other, but two draws run by one agency for the same tag are one good, so an allocator cannot shed its obligation by splitting a practice. The agency matters because another allocator has none of its units to divide. This distinguishes the thesis from luck egalitarianism (Dworkin 1981; Cohen 1989; Temkin

[6] 19.31.17.9 NMAC (2025).

1993), whose currency is compensation, so that a shortfall in one good can be made up in another; mine is silent on whether the loser is worse off overall.

Memory is bound to the claim. The corrective claim is a strengthened claim to the good, not a credit. Someone who applied unsuccessfully for ten years and then gave up hunting is owed nothing; there is no tag she has a claim to. Memory is not compensation: shortfall arises under the fairest rule, since the residual lottery produces it, so correcting it presupposes no wrong. A boundary question follows. *Transition*: when a forgetting practice adopts memory, the whole record counts, wins and losses, since the shortfall existed whether or not the rule tracked it; a rectificatory view would count only the losses.

7 Objections

Losing on purpose. That memory rewards losing is a familiar objection: the NBA moved from a worst-first draft order to a lottery because of the incentive to lose (Elster 1988, 123 n. 46). In my cases losing is not under the claimant's control: she can only apply, and applying without a genuine claim is the problem of verifying claims, which every allocation faces. Where losing is controllable, as in a league, fairness is in tension with incentive, a reason to weigh fairness against other things, not evidence that fairness requires forgetting. Memory does raise the value of a manufactured claim, a ticket that grows; the response is to attach it to participation that costs something, as a paid licence does: a constraint on implementing fairness, not evidence that fairness has no memory.

Equality, not fairness. Someone might grant the verdict and deny that fairness is what condemns the outcomes: what is bad is inequality, or that some are badly off (Parfit 2000). Suppose the persistent loser is, all things considered, no worse off than the repeat winner. Her complaint survives: this agency, dividing this good among equal claims, has given her less than her share. That is comparative, concerns the good in question, and is addressed to the allocator, the marks of a fairness complaint in Broome's sense; a priority view cannot explain why it lapses when she stops claiming.

Ex ante. The last objection comes from the view, descending from Diamond's (1967) reply to Harsanyi (1955) (Frick 2015; Otsuka and Voorhoeve 2009), that a procedure's fairness is a matter of the prospects it gives people at the time of choice. On that view the forgetting rule and rotation are equally fair, since each claimant starts with the same prospects under both; the forgetting rule's defect, as the first row of Table 1 showed, is entirely ex post. First reply: the ex ante view must say when the ex ante is. At the beginning of the practice, forgetting and rotation are identical, and the Division Principle settles what the view leaves open; at the beginning of each period, the view asserts separability with respect to history-blind claims, which the cake refutes. Second, my view is ex ante exactly where the ex ante view is right: a claimant is owed chances insofar as division is unavailable to her and units insofar as it is, which is why the surrogate constraint conditions on her tenure, which settles whether division was available to her, and not on her draws, which are the surrogate itself. Dated at her entry, every rule gives her the same prospects, by conservation; the ex ante view, built for one-shot cases where division never is, can tell none of them apart.

8 Conclusion

Fairness has memory where the good recurs. What the memory tracks is a claimant's shortfall from her share of the good being divided; it is owed by the agency dividing the good, for as long as she has a claim to it; and it may be pressed only through a rule whose transfers between classes are a by-product of a winner's not being taxable rather than its mechanism, and no harder than excludes a claimant when she holds a whole unit of another's share. Where

winners leave and some will leave unserved, nobody present holds a unit, so any weight on waiting is transfer and nothing else, and the same constraint forbids memory. The result was visible in Broome's own illustration of the Division Principle: military service, divided by conscripting everyone for a short time. Everyone's short time is a rotation. The lottery is what one does when one cannot rotate, and one can rotate more often than its defenders have noticed.

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